

Lab 2 Activity

PSYC 7804 - Spring 2026

Today we will be looking at the `insurance.csv` dataset. These data are data simulated according to the US census data and can be found [here on this Kaggle page](#). To load the data run the following code:

```
dat <- rio::import("https://fabio-setti.netlify.app/data/insurance.csv")
```

Here is a descriptions of the variables in the data:

variable	description
age	Age of primary beneficiary
sex	Insurance contractor gender, female, male
bmi	Body mass index
children	Number of children covered by health insurance / Number of dependents
smoker	Whether the beneficiary is a smoker
region	The beneficiary's residential area in the US, northeast, southeast, southwest, northwest
charges	Individual medical costs billed yearly by health insurance in \$

1. Create a scatterplot with **bmi** on the x -axis and **charges** on the y -axis and draw a regression line. Does the regression line match your expectations? Do you see something a bit strange about the distribution of the dots around the regression line?
2. Create Mean-centered and Standardized versions of both **bmi** and **charges**.
 - What is the mean, standard deviation, and range (you can use the **range()** function) of **bmi** and **charges** and their linearly transformed counterparts?
3. Run an unstandardized linear regression with **bmi** predicting **charges**. Write the regression equation and interpret the intercept and the slope.
 - Do you notice anything off with the interpretation of the intercept?
4. Using any of either the original or linearly transformed variables, run a regression such that the intercept represents the expected value of **charges** when **bmi** is at the sample mean. According to these data, what are the expected annual **charges** in dollars for an individual with average BMI?
5. Using any of either the original or linearly transformed variables, find the correlation between **charges** and **bmi**. (only use the **lm()** and **summary()** function to answer this question)
6. Assuming that some variable is roughly normally distributed, standardization transforms the variable into Z-scores (see [here](#)). In the case of a Z-score, we know that a value of 0 generally represents the 50th percentile, while a value of 1 (i.e., *1 standard deviation above the mean*) represents the 84th percentile. The **pnorm()** function returns the corresponding percentile given any Z-score:

```
pnorm(0)
```

```
[1] 0.5
```

```
pnorm(1)
```

```
[1] 0.8413447
```

Assuming that **charges** is approximately normally distributed, in what percentile of **charges** are individuals who are 1.5 standard deviations above the mean **bmi** predicted to be?

- Think about the effect of **bmi** on **charges**. Assuming a roughly normal distribution, 1.5 corresponds to what **bmi** percentile? Given the magnitude of the change in **bmi** necessary for the predicted percentile of **charges**, how impactful do you think is the effect of **bmi** on **charges**?